Note on Bandit Algorithms

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1 Introduction

Definition.

- Actions : $A_1, A_2, \cdots \in \mathcal{A}$ and we call $\mathcal{A}$ the action set.
- Rewards : $X_1, X_2, \cdots \in \mathcal{R}$.
- History : $H_{t-1} = (A_1, X_1, \cdots, A_{t-1}, X_{t-1})$.
- Policy : $\pi \in \Pi : H_{t-1} \mapsto A_t$ and we call Π the competitor class.
- Environment: $\mathcal{E} : A_t \mapsto X_t$.
- Regret of a learner relative to a policy π until $n \in \mathbb{N}$ is

$$R_n = \mathbb{E}\left[\sum_{t=1}^{n} \mathcal{E}(\pi(H_{t-1})) - \sum_{t=1}^{n} X_t\right].$$

  where H_0 represents the initial condition.

Example A. n environment is called a **stochastic Bernoulli bandit** if

- $\mathcal{A} = \{1, 2, \ldots, k\}$,
- $X_t \in \{0, 1\}$,
- $^{\forall} a \in \mathcal{A}$, $\exists\, \mu_a \in [0,1]^k$ such that $\mathbb{P}\left[X_t = 1 | A_t = a\right] = \mu_a$.

Here the optimal policy is to play fixed action $a^* = \underset{a \in \mathcal{A}}{\operatorname{argmax}}\, \mu_a$. That is, the competitor class Π is the set of all constant policies $\Pi = \{\pi_1, \ldots, \pi_k\}$ where π_i chooses action i in every round. The regret relative to the optimal policy π^* until n rounds is

$$\begin{aligned} R_n &= \mathbb{E}\left[\sum_{t=1}^{n} \mathcal{E}(\pi^*(H_{t-1})) - \sum_{t=1}^{n} X_t\right] \\ &= n \max_{a \in \mathcal{A}} \mu_a - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right]. \end{aligned}$$

Definition. Stochastic stationary bandits are the environments restricted to generate the rewards in response to each action from a distribution independent of the previous history.

Example Linear Stochastic Bandits.

- $\mathcal{A} \subset \mathbb{R}^d$,
- $X_t = \langle a, \theta \rangle + \eta_t$ for $\theta \in \mathbb{R}^d, a \in \mathcal{A}$ and η_t is a standard Gaussian.

2 Foundations of Probability

Definition. Let an **event set** Ω be given.

- A set $\mathcal{F} \subset 2^\Omega$ is called a **σ-algebra** on Ω if
  (1) $\Omega \in \mathcal{F}$,
  (2) $A \in \mathcal{F} \implies A^c \in \mathcal{F}$,
  (3) $A_1, A_2, \ldots \in \mathcal{F} \implies \bigcup_{i=1}^\infty A_i \in \mathcal{F}$.
- $\mathcal{G} \subseteq \mathcal{F}$ is a **sub-σ-algebra** if $\mathcal{G}$ is a σ-algebra on Ω.
- $(\Omega, \mathcal{F})$ is called a **measurable space** if $\mathcal{F}$ is a σ-algebra on Ω.
- $\mathcal{B}(\mathbb{R})$ is the **Borel σ-algebra** on $\mathbb{R}$, which is the smallest σ-algebra containing all open intervals (a, b) for $a, b \in \mathbb{R}$.

Definition. Let a measurable space $(\Omega, \mathcal{F})$ be given.

- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is **σ-additive** if
  (1) $\mu(\emptyset) = 0$,
  (2) $A_1, A_2, \ldots \in \mathcal{F}$ are disjoint $\implies \mu\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty \mu(A_i)$.
- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is a **measure** if μ is σ-additive.
- $(\Omega, \mathcal{F}, \mu)$ is called a **measure space** if μ is a measure on $(\Omega, \mathcal{F})$.
- A measure μ is a **probability measure** if $\mu(\Omega) = 1$.
- $(\Omega, \mathcal{F}, \mu)$ is called a **probability space** if μ is a probability measure on $(\Omega, \mathcal{F})$.
- A function $X : \Omega \mapsto \mathbb{R}$ is called a **measurable function** if $^\forall B \in \mathcal{B}(\mathbb{R})$, $X^{-1}(B) \in \mathcal{F}$.
- A **filtration** is a sequence $(\mathcal{F}_t)_{t=0}^n$ of sub-σ-algebras of $\mathcal{F}$ where $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$ for all $t < n$.

Definition. Let a measure space $(\Omega, \mathcal{F}, \mu)$ be given.

- A function $f : \Omega \mapsto [0, \infty]$ is called a **simple function** if $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ for some $a_1, a_2, \ldots, a_n \in [0, \infty]$ and $A_1, A_2, \ldots, A_n \in \mathcal{F}$.
- The **integral of a simple function** $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\int_\Omega f d\mu := \sum_{i=1}^n a_i \mu(A_i)$.
- The **integral of a non-negative measurable function** $f : \Omega \mapsto [0, \infty]$ is defined as $\int_\Omega f d\mu := \sup_g \int_\Omega g d\mu$ where the supremum is taken over all simple functions g such that $0 \leq g(\omega) \leq f(\omega)$ for all $\omega \in \Omega$.
- The **integral of a measurable function** $f : \Omega \mapsto \mathbb{R}$ is defined as $\int_\Omega f d\mu := \int_\Omega f^+ d\mu - \int_\Omega f^- d\mu$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.
- A function $f : \Omega \mapsto \mathbb{R}$ is called **integrable** if $\int_\Omega |f| d\mu < \infty$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ be given.

- A measurable function $X : \Omega \mapsto \mathbb{R}$ is called a **random variable**.
- The **conditional probability** $\mathbb{P}[A|B] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}$ for $A, B \in \mathcal{F}$ and $\mathbb{P}[B] > 0$.
- Two events $A, B \in \mathcal{F}$ are called **independent** if $\mathbb{P}[A \cap B] = \mathbb{P}[A] \cdot \mathbb{P}[B]$.
- Two random variables X, Y are called **independent** if $^\forall A, B \in \mathcal{B}(\mathbb{R})$, $\mathbb{P}[X \in A, Y \in B] = \mathbb{P}[X \in A] \cdot \mathbb{P}[Y \in B]$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X be given.

- The **expectation** of a random variable X is defined as $\mathbb{E}[X] := \int_\Omega X d\mathbb{P}$.
- For a sub-σ-algebra $\mathcal{H} \subseteq \mathcal{F}$, the **conditional expectation** of X given $\mathcal{H}$ is denoted by $\mathbb{E}[X|\mathcal{H}]$ and is defined as a random variable g on Ω such that for all $H \in \mathcal{H}$,

$$\int_H g \, d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem 2.11. *If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and X is a integrable random variable and $\mathcal{H} \subseteq \mathcal{F}$ is a sub-σ-algebra, there exists a unique (almost surely) random variable $\mathbb{E}[X|\mathcal{H}]$ such that for all $H \in \mathcal{H}$,*

$$\int_H \mathbb{E}[X|\mathcal{H}] \, d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem 2.12. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G}, \mathcal{G}_1, \mathcal{G}_2 \subseteq \mathcal{F}$ be sub-σ-algebras of $\mathcal{F}$, X, Y be integrable random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then the following properties hold:*

(1) If $X \geq 0$ almost surely, then $\mathbb{E}[X|\mathcal{G}] \geq 0$ almost surely.

(2) $\mathbb{E}[1|\mathcal{G}] = 1$ almost surely.

(3) $\mathbb{E}[X + Y|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}] + \mathbb{E}[Y|\mathcal{G}]$ almost surely.

(4) If Y is $\mathcal{G}$-measurable and $\mathbb{E}[XY] < \infty$, then $\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}]$ almost surely.

(5) If $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then $\mathbb{E}[X|\mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1]$ almost surely.

3 Stochastic Processes and Markov Chains

Definition. Let two measurable spaces $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be given.

- $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ are called **isomorphic** if there exists a bijective function $f : \mathcal{X} \mapsto \mathcal{Y}$ such that $A \in \mathcal{F} \iff f(A) \in \mathcal{G}$.
- $(\mathcal{X}, \mathcal{F})$ is a **Borel space** if it is isomorphic to $(\mathbb{R}, \mathcal{B}(A))$ for some set $A \in \mathcal{B}(\mathbb{R})$.

Theorem 3.1. *Let μ be a probability measure on a Borel space S and λ be the Lebesgue measure on $([0,1], \mathcal{B}([0,1]))$. Then there exists a sequence of independent random element $X_1, X_2, \ldots$ on $([0,1], \mathcal{B}([0,1]), \lambda)$ such that $\lambda_{X_t}(\cdot) := \lambda(X_t^{-1}(\cdot)) = \mu(\cdot)$ for all t.*

Definition. For "countable" set $\mathcal{T}$, A stochastic process is a collection of random variables $X_t : t \in \mathcal{T}$.

Theorem 3.2. *Let Borel spaces $(\Omega_n, \mathcal{F}_n)$ and measures μ_n on $(\Omega_1 \times \cdots \times \Omega_n, \mathcal{F}_1 \otimes \cdots \otimes \mathcal{F}_n)$ are given for each $n \in \mathbb{N}$.*

$$\begin{aligned} &{}^\forall n \in \mathbb{N} \text{ and } A \in \mathcal{F}_1 \otimes \cdots \otimes \mathcal{F}_n, \quad \mu_{n+1}(A \times \Omega_{n+1}) = \mu_n(A) \\ &\implies \exists \text{ a probability space } (\Omega, \mathcal{F}, \mathbb{P}) \\ &\text{and random elements } X_1, X_2, \ldots \text{ with } X_t : \Omega \to \Omega_t \text{ such that } \mathbb{P}_{X_1,\ldots,X_n} = \mu_n \text{ for all } n. \end{aligned}$$

Definition. Let $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be measurable spaces. A **probability kernel** or **Markov kernel** form $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y}, \mathcal{G})$ is a function $K : \mathcal{X} \times \mathcal{G} \to [0,1]$ such that

(a) $K(x, \cdot)$ is a measure for all $x \in \mathcal{X}$ and

(b) $K(\cdot, A)$ is $\mathcal{F}$-measurable for all $A \in \mathcal{G}$.

Notation: $K_x(A)$ or $K(A|x)$

Definition. Let K_1 be a probability kernel from $(\mathcal{X}, \mathcal{F}) \to (\mathcal{Y}, \mathcal{G})$ and K_2 be a probability kernel from $(\mathcal{Y}, \mathcal{G}) \to (\mathcal{Z}, \mathcal{H})$. Then the product kernel $K_1 \otimes K_2$ is the probability kernel from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y} \times \mathcal{Z}, \mathcal{G} \otimes \mathcal{H})$ defined by

$$(K_1 \otimes K_2)(x, A) = \int_{\mathcal{Y}} \int_{\mathcal{Z}} \mathbb{1}_A((y, z)) K_2(y, dz) K_1(x, dy).$$

Theorem 3.3. *Let $(\Omega_n, \mathcal{F}_n)_{n=1}^{\infty}$ be a sequence of measurable spaces.*

K_1 : a probability measure on $(\Omega_1, \mathcal{F}_1)$,

K_n : a probability kernel from $\Pi_{t=1}^{n-1} \Omega_t$ to Ω_n for $n \geq 2$.

$\implies \exists$ a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

and random elements $(X_t)_{t=1}^{\infty}$ with $X_t : \Omega \to \Omega_t$ such that $\mathbb{P}_{X_1,\ldots,X_n} = \bigotimes_{t=1}^{n} K_t$ for all n.

Definition. A **homogeneous Markov chain** is a sequence of random elements $(X_t)_{t=1}^{\infty}$ taking values in **state space** $\mathcal{S} = (\mathcal{X}, \mathcal{F})$ and with

$$\mathbb{P}(X_{t+1} \in \cdot \,| X_1, \ldots, X_t) = \mathbb{P}(X_{t+1} \in \cdot \,| X_t) = \mu(X_t, \cdot), \qquad \text{a.s.}$$

Here μ is a probability kernel from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{X}, \mathcal{F})$.

Definition 3.4. A $\mathbb{F}$-adapted sequence of random variables $(X_t)_{t \in \mathbb{N}_+}$(i.e. X_t is $\mathcal{F}_t$-measurable for all $t \in \mathbb{N}$) is a $\mathbb{F}$-adapted **martingale** if

(a) $\mathbb{E}[|X_t|] < \infty$ for all t.

(b) $\mathbb{E}[X_{t+1}|\mathcal{F}_t] = X_t$ for all t.

Supermartingale if less-than, submartingale if greater-than.

Example 3.5. $(Y_t)_{t \in \mathbb{N}}$ be a sequence of independent random variables with $\mathbb{P}(Y_t = 1) = \mathbb{P}(Y_t = -1) = \frac{1}{2}$. Then the winnings after t rounds is $S_t = \sum_{i=1}^{t} Y_i$, a martingale adapted to the filtration $(\mathcal{F}_t)_{t \in \mathbb{N}_+}$, where $\mathcal{F}_t = \sigma(Y_1, \ldots, Y_t)$.

Definition 3.6. Let $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{N}}$ be a filtration. A random variable $\tau \in \mathbb{N} \cup \{\infty\}$ is a **stopping time** with respect to the filtration $\mathbb{F}$ if

$$\{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in \mathbb{N}.$$

The σ-algebra generated by the stopping time τ is

$$\mathcal{F}_\tau = \{A \in \mathcal{F}_\infty : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in \mathbb{N}\}.$$

Theorem 3.8 (Doob's Optional Stopping Theorem)**.** *Let $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{N}}$ be a filtration and $(X_t)_{t \in \mathbb{N}}$ be an $\mathbb{F}$-adapted martingale. Let τ be a stopping time with respect to $\mathbb{F}$, and one of the following conditions is satisfied:*

(a) There exists an $n \in \mathbb{N}$ such that $\mathbb{P}(\tau > n) = 0$.

(b) $\mathbb{E}[\tau] < \infty$ and there exists a constant $c \in \mathbb{R}$ such that $|X_{t+1} - X_t| \leq c$ for all t, $\mathbb{E}[|X_{t+1} - X_t||\mathcal{F}_t] \leq c$ a.s. for $\tau > t$.

(c) There exists a constant c such that $|X_{t \wedge \tau}| \leq c$ a.s. for all $t \in \mathbb{N}$.

Then X_τ is almost surely well-defined and

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0].$$

Theorem 3.9 (Maximal inequality)**.** *Let $(X_t)_{t=0}^{\infty}$ be a supermartingale with $X_t \geq 0$ a.s. for all $t \in \mathbb{N}$. Then for any $\epsilon > 0$,*

$$\mathbb{P}\left(\sup_{t \in \mathbb{N}} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_0]}{\epsilon}.$$

Theorem 3.10. *Let $(X_t)_{t=0}^{\infty}$ be a submartingale with $X_t \geq 0$ a.s. for all $t \in \mathbb{N}$. Then for any $\epsilon > 0$,*

$$\mathbb{P}\left(\max_{t \in 0,1,\ldots,n} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_n]}{\epsilon}.$$

4 Stochastic Bandits

Definition. A **stochastic bandit** is a collection of distributions $\nu = \{P_a : a \in \mathcal{A}\}$.

Here we are challenging to maximize the total reward $S_n = \sum_{t=1}^n X_t$ over n rounds.

Definition. A set of bandits $\mathcal{E}$ for which $\nu \in \mathcal{E}$ is called the environment class.

- An environment class $\mathcal{E}$ is **unstructured** if $\mathcal{A}$ is finite and

  $$^\forall a \in \mathcal{A}, \exists \text{ sets of distributions } \mathcal{M}_a \text{ such that } \mathcal{E} = \{\nu = (P_a : a \in \mathcal{A}) : P_a \in \mathcal{M}_a, {}^\forall a \in \mathcal{A}\}.$$

  or, in short, $\mathcal{E} = \times_{a \in \mathcal{A}} \mathcal{M}_a$.
- Environment classes with finite number of degrees of freedom are called **parametric** (Bernoulli).
- Environment classes that are not parametric are called **nonparametric** (subgaussian).
- An environment class is called **structured** if it is not unstructured.

Example 4.1. Let $\mathcal{A} = \{1, 2\}$ and $\mathcal{E} = \{(\mathcal{B}(\theta), \mathcal{B}(1-\theta)) : \theta \in [0, 1]\}$ where $\mathcal{B}(\theta)$ denotes the Bernoulli distribution with mean θ. Then $\mathcal{E}$ is a structured parametric environment class.

Example 4.2 (Stochastic linear bandit)**.** Let $\mathcal{A} \subseteq \mathbb{R}^d$ and $\theta \in \mathbb{R}^d$ and

$$\nu_\theta = (\mathcal{N}(\langle a, \theta \rangle, 1) : a \in \mathcal{A}) \qquad \text{and} \quad \mathcal{E} = \{\nu_\theta : \theta \in \mathbb{R}^d\}.$$

Example 4.3. Consider an undirected graph G with vertices $V = 1, \ldots, |V|$ and edges $E = 1, \ldots, |E|$. In each round the learner chooses a path from vertex 1 to vertex $|V|$. Then each edge $e \in [E]$ is removed from the graph with probability $1 - \theta_e$ for unknown $\theta \in [0, 1]^{|E|}$. In this problem, $\mathcal{A}$ is the set of paths and

$$\nu_\theta = \left(\mathcal{B}\left(\prod_{e \in a} \theta_e \right) : a \in \mathcal{A} \right) \qquad \text{and} \quad \mathcal{E} = \{\nu_\theta : \theta \in [0, 1]^{|E|}\}.$$

Definition. Let $\nu = (P_a : a \in \mathcal{A})$ be a stochastic bandit.

$$\mu_a(\nu) := \int_{-\infty}^{\infty} x \, dP_a(x)$$

Also let $\mu^*(\nu) := \max_{a \in \mathcal{A}} \mu_a(\nu)$, the largest mean of all the arms.

Definition. The **regret** of policy π on bandit instance ν is

$$R_n(\pi, \nu) := n\mu^*(\nu) - \mathbb{E}\left[\sum_{t=1}^n X_t \right]$$

where the expectation is taken with respect to the probability measure on outcomes induced by the interaction of π and ν.

Lemma 4.4. *Let ν be a stochastic bandit environment. Then*

- $R_n(\pi, \nu) \geq 0$ *for all policies π,*
- *the policy π choosing $A_t \in \arg\max_a \mu_a(\nu)$ for all t satisfies $R_n(\pi, \nu) = 0$, and*
- *if $R_n(\pi, \nu) = 0$ for some policy π, then $\mathbb{P}[\mu_{A_t}(\nu) = \mu^*(\nu)] = 1$ for all $t \in [n]$.*

Proof. (a) Since $\mu^*(\nu) = \max_{a \in \mathcal{A}} \mu_a(\nu)$,

$$n\mu^*(\nu) \geq \mathbb{E}\left[\sum_{t=1}^{n} \mu_{A_t}(\nu)\right].$$

By the linearity of expectation,

$$\mathbb{E}\left[\sum_{t=1}^{n} \mu_{A_t}(\nu)\right] = \sum_{t=1}^{n} \mathbb{E}\left[\mu_{A_t}(\nu)\right].$$

By the stochastic bandit assumption and the tower property,

$$\begin{aligned}
\sum_{t=1}^{n} \mathbb{E}\left[\mu_{A_t}(\nu)\right] &= \sum_{t=1}^{n} \mathbb{E}\left[\mathbb{E}\left[X_t \mid H_{t-1}, A_t\right]\right] \\
&= \sum_{t=1}^{n} \mathbb{E}\left[X_t\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{n} X_t\right],
\end{aligned}$$

using linearity of expectation again. Therefore,

$$R_n(\pi, \nu) = n\mu^*(\nu) - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right] \geq 0.$$

(b) Let π be the policy choosing $A_t \in \arg\max_a \mu_a(\nu)$ for all t. Then $\mu^*(\nu) = \mu_{A_t}(\nu)$ for all t, which implies

$$\begin{aligned}
R_n(\pi, \nu) &= n\mu^*(\nu) - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right] \\
&= \sum_{t=1}^{n} \left(\mu^*(\nu) - \mathbb{E}\left[\mu_{A_t}(\nu)\right]\right) \\
&= \mathbb{E}\left[\sum_{t=1}^{n} \left(\mu^*(\nu) - \mu_{A_t}(\nu)\right)\right] \\
&= 0
\end{aligned}$$

and therefore $R_n(\pi, \nu) = 0$.

(c) Suppose $R_n(\pi, \nu) = 0$ for some policy π. Then

$$\sum_{t=1}^{n} \mathbb{E}\left[\mu^*(\nu) - \mu_{A_t}(\nu)\right] = 0.$$

Since $\mu^*(\nu) - \mu_{A_t}(\nu) \geq 0$ for all t, $\mu^*(\nu) = \mu_{A_t}(\nu)$ almost surely for all $t \in \{1, \ldots, n\}$.

□

Remark (Factorisation of the regret)**.** The learner cannot always achieve zero regret, so the relatively weak objective is to find a policy π with

$$^{\forall}\, \nu \in \mathcal{E}, \quad \lim_{n \to \infty} \frac{R_n(\pi, \nu)}{n} = 0.$$

Or further objective, finding some constants $C > 0$ and $p < 1$ such that

$$^{\forall}\, \nu \in \mathcal{E}, \quad R_n(\pi, \nu) \leq Cn^p.$$

Another alternative is to find a function $C : \mathcal{E} \to [0, \infty)$ and $f : \mathbb{N} \to [0, \infty)$ such that

$$^{\forall}\, \nu \in \mathcal{E}, \quad R_n(\pi, \nu) \leq C(\nu) f(n).$$

Bayesian regret is another candidate objective. For prior probability measure Q on $\mathcal{E}$, the Bayesian regret is the average of the regret with respect to the prior Q, i.e.,

$$\mathrm{BR}_n(\pi, Q) = \int_{\mathcal{E}} R_n(\pi, \nu)\, dQ(\nu)$$

where the regret R_n is measurable.

Definition. Let $\nu = (P_a : a \in \mathcal{A})$ be a stochastic bandit. Then the **suboptimality gap** of action a is defined by

$$\Delta_a(\nu) := \mu^*(\nu) - \mu_a(\nu).$$

Also let

$$T_a(t) := \sum_{s=1}^{t} \mathbb{1}\{A_s = a\}$$

be the number of times action a has been chosen up to time t.

Lemma 4.5 (Regret decomposition lemma)**.** *For any policy π and stochastic bandit ν with $\mathcal{A}$ finite or countable and horizon $n \in \mathbb{N}$, the regret R_n of policy π in ν satisfies*

$$R_n(\pi, \nu) = \sum_{a \in \mathcal{A}} \Delta_a(\nu) \mathbb{E}[T_a(n)].$$

Proof. For any $t \in \mathbb{N}$, $\sum_{a \in \mathcal{A}} \mathbb{1}\{A_t = a\} = 1$. Hence $S_n = \sum_t X_t = \sum_t \sum_a X_t \mathbb{1}\{A_t = a\}$. That is

$$R_n = n\mu^* - \mathbb{E}[S_n] = \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\}].$$

Since $\mathbb{E}[X_t \mid H_{t-1}, A_t] = \mu_{A_t}$, we have

$$\begin{aligned}
\mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\} \mid A_t] &= \mathbb{1}\{A_t = a\}\mathbb{E}[\mu^* - X_t \mid A_t] \\
&= \mathbb{1}\{A_t = a\}(\mu^* - \mu_a) \\
&= \mathbb{1}\{A_t = a\}\Delta_a.
\end{aligned}$$

Therefore

$$\begin{aligned}
R_n &= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\}] \\
&= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[\mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\} \mid A_t]] \\
&= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[\Delta_a \mathbb{1}\{A_t = a\}] \\
&= \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)].
\end{aligned}$$

□

Exercise 4.10. Suppose ν is a finite-armed stochastic bandit and π is a polict such that

$$\lim_{n \to \infty} \frac{R_n(\pi, \nu)}{n} = 0.$$

Let $T^*(n) = \sum_{t=1}^{n} \mathbb{1}\{\mu_{A_t} = \mu^*\}$ be the number of times an optimal arm is chosen. Prove or diprove following statement:

(a) $\lim_{n\to\infty} \mathbb{E}[T^*(n)]/n = 1$.

(b) $\lim_{n\to\infty} \mathbb{P}(\Delta_{A_t} > 0) = 0$.

Solution. (a) True, the following are the proof;

Proof. If every arm is optimal, then $\lim_{n\to\infty} \frac{\mathbb{E}[T^*(n)]}{n} = \lim_{n\to\infty} \frac{n}{n} = 1$. Suppose not. That is,

$$\exists\, a_0 \in \operatorname*{argmin}_{a\in\{b\in\mathcal{A}:\mu_b<\mu^*\}} \{\mu^* - \mu_a\}.$$

Letting $c = \mu^* - \mu_{a_0}$ follows that $\mathbb{E}[(\mu^* - \mu_{A_t})] \geq c \cdot \mathbb{P}[\mu^* \neq \mu_{A_t}]$. Hence

$$\begin{aligned}\frac{R_n}{n} &= \sum_{t=1}^{n} \frac{\mathbb{E}[\mu^* - \mu_{A_t}]}{n} \\ &\geq c\frac{\sum_{t=1}^{n} \mathbb{P}[\mu^* \neq \mu_{A_t}]}{n}.\end{aligned}$$

$$\therefore \lim_{n\to\infty} \frac{\sum_{t=1}^{n} \mathbb{P}[\mu^* \neq \mu_{A_t}]}{n} = 0.$$

Hence $^\forall n \in \mathbb{N}$,

$$\begin{aligned}\frac{\mathbb{E}[T^*(n)]}{n} &= \sum_{t=1}^{n} \frac{\mathbb{E}[\mathbb{1}_{\mu_{A_t}=\mu^*}]}{n} \\ &= \frac{\sum_{t=1}^{n} \mathbb{P}[\mu_{A_t} = \mu^*]}{n} \\ &= 1 - \frac{\sum_{t=1}^{n} \mathbb{P}[\mu_{A_t} \neq \mu^*]}{n}.\end{aligned}$$

$$\lim_{n\to\infty} \frac{\mathbb{E}[T^*(n)]}{n} = 1 - \lim_{n\to\infty} \frac{\sum_{t=1}^{n} \mathbb{P}[\mu_{A_t} \neq \mu^*]}{n} = 1.$$

□

(b) False, the following is a counterexample;

Consider $\mathcal{A} = \{a_0, a_1\}$ with $\mu^* = \mu_{a_0}$ and $\mu^* - \mu_{a_1} = \Delta_{a_1} > 0$. Let the policy π choose a_1 at 2^k rounds for $k \in \mathbb{N}$. Then

$$\begin{aligned}\lim_{n\to\infty} \frac{R_n(\pi)}{n} &= \lim_{n\to\infty} \frac{\sum_{t=1}^{n} \mathbb{E}[\mu^* - \mu_{A_t}]}{n} \\ &\leq \lim_{n\to\infty} \frac{\sum_{t=1}^{\lfloor \log_2 n + 100 \rfloor} \Delta_{a_1}}{n} \\ &= \lim_{n\to\infty} \frac{\Delta_{a_1} \lfloor \log_2 n + 100 \rfloor}{n} \\ &= 0.\end{aligned}$$

But $\lim_{n\to\infty} \mathbb{P}[\Delta_{A_{2^k}} > 0] = 1$ for $k \in \mathbb{N}$, which disproves (b).

Exercise 4.11 (Retirement policy)**.** Let $\mathcal{M}_1$ be a set of distributions on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, with finite means and $\mathcal{M}_2 = \{\delta_{\mu_2}\}$ be a singleton set containing the Dirac measure at μ_2. Construct the set of bandits $\mathcal{E} = \mathcal{M}_1 \times \mathcal{M}_2$. Here a policy $\pi = (\pi_t)_{t\in\mathbb{N}}$ is called a retirement policy if $a_t = 2$, then

$$\pi_{t+1}(2 \mid a_1, x_1, \ldots, a_t, x_t) = 1, \quad {}^\forall (a_s)_{s=1}^{t-1} \text{ and } (x_s)_{s=1}^{t}.$$

(a) Let n be fixed and $\pi = (\pi_t)_{t=1}^n$ be any policy. Prove there exists a retirement policy $\pi' = (\pi'_t)_{t=1}^n$ such that for all $\nu \in \mathcal{E}$,

$$R_n(\pi', \nu) \leq R_n(\pi, \nu)$$

(b) Let $\mathcal{M}_1 = \{\mathcal{B}(\mu_1) : \mu_1 \in [0,1]\}$ and suppose that $\pi = (\pi_t)_{t=1}^{\infty}$ is a retirement policy. Prove there exists a bandit $\nu \in \mathcal{E}$ such that

$$\limsup_{n\to\infty} \frac{R_n(\pi,\nu)}{n} > 0.$$

Proof. (a) Let

$$\mathcal{H}_j := (\{1,2\} \times \mathbb{R})^j, \qquad j \in \{0,\dots,n\},$$

where $\mathcal{H}_0 = \{\varnothing\}$. Fix $n \in \mathbb{N}$ and a policy $\pi = (\pi_j)_{j=1}^n$. Let $U_1,\dots,U_n \overset{\text{i.i.d.}}{\sim} \text{Unif}(0,1)$. For $j \in \{0,\dots,n-1\}$ and $h \in \mathcal{H}_j$, define

$$\psi_{j+1}(h) := \begin{cases} 1, & U_{j+1} \le \pi_{j+1}(1 \mid h), \\ 2, & U_{j+1} > \pi_{j+1}(1 \mid h). \end{cases}$$

Define

$$\Gamma : \bigcup_{j=0}^{n} \mathcal{H}_j \longrightarrow \bigcup_{j=0}^{n} \mathcal{H}_j$$

such that $^\forall j \in \{0,\dots,n\}$ and $^\forall h \in \mathcal{H}_j$,

$$\Gamma(h) := \begin{cases} h, & j = n, \\ h, & j < n \text{ and } \psi_{j+1}(h) = 1, \\ \Gamma\big(h,(2,\mu_2)\big), & j < n \text{ and } \psi_{j+1}(h) = 2. \end{cases}$$

Now fix $\nu = (\nu_1, \delta_{\mu_2}) \in \mathcal{E}$. Let

$$Y_1, Y_2, \dots \overset{\text{i.i.d.}}{\sim} \nu_1$$

be the rewards obtained from arm 1. Define $\widetilde{H}_0,\dots,\widetilde{H}_n$ and $A'_1,\dots,A'_n$ by

$$\widetilde{H}_0 := \Gamma(\varnothing),$$

and, for $t \in \{1,\dots,n\}$,

$$A'_t := \begin{cases} 1, & |\widetilde{H}_{t-1}| < n, \\ 2, & |\widetilde{H}_{t-1}| = n, \end{cases}$$

and

$$\widetilde{H}_t := \begin{cases} \Gamma\big(\widetilde{H}_{t-1},(1,Y_t)\big), & |\widetilde{H}_{t-1}| < n, \\ \widetilde{H}_{t-1}, & |\widetilde{H}_{t-1}| = n. \end{cases}$$

Then $^\forall t \in \{0,\dots,n\}, \widetilde{H}_t \in \bigcup_{j=0}^n \mathcal{H}_j$. Define $^\forall h \in \mathcal{H}_{t-1}$ and $^\forall a \in \{1,2\}$,

$$H'_{t-1} \in \mathcal{H}_{t-1} \text{ and } \pi'_t(a \mid h) = \mathbb{P}(A'_t = a \mid H'_{t-1} = h).$$

for all $t \in \{1,\dots,n\}$. We first show that π' is a retirement policy. Suppose that $A'_{t_0} = 2$ for some $t_0 \in \{1,\dots,n\}$. Then

$$|\widetilde{H}_{t_0-1}| = n.$$

Hence, by the definition of $\widetilde{H}_t$,

$$\widetilde{H}_s = \widetilde{H}_{t_0-1} \qquad \text{for all } s \ge t_0 - 1.$$

Therefore,

$$A'_s = 2 \qquad \text{for all } s \ge t_0.$$

Thus, π' is a retirement policy. Let

$$K := T_1^{\pi'}(n) = \sum_{t=1}^{n} \mathbb{1}_{\{A'_t = 1\}}.$$

By construction,

$$\widetilde{H}_K \in \mathcal{H}_n.$$

Moreover, conditionally on each prefix of $\widetilde{H}_K$, its next action is chosen according to the same conditional distribution as under π, and its next reward has distribution ν_1 if that action is 1, and equals μ_2 if that action is 2. Consequently,

$$\widetilde{H}_K \overset{d}{=} H_n^{\pi},$$

where H_n^{π} denotes the n-round history generated by π in the bandit ν. Since K is exactly the number of occurrences of action 1 in $\widetilde{H}_K$, it follows that

$$T_1^{\pi'}(n) \overset{d}{=} T_1^{\pi}(n).$$

Hence,

$$\mathbb{E}_{\nu}^{\pi'}[T_1^{\pi'}(n)] = \mathbb{E}_{\nu}^{\pi}[T_1^{\pi}(n)].$$

Since $T_2(n) = n - T_1(n)$,

$$\mathbb{E}_{\nu}^{\pi'}[T_2^{\pi'}(n)] = \mathbb{E}_{\nu}^{\pi}[T_2^{\pi}(n)].$$

Finally, by the regret decomposition lemma,

$$\begin{aligned} R_n(\pi', \nu) &= \Delta_1(\nu)\mathbb{E}_{\nu}^{\pi'}[T_1^{\pi'}(n)] + \Delta_2(\nu)\mathbb{E}_{\nu}^{\pi'}[T_2^{\pi'}(n)] \\ &= \Delta_1(\nu)\mathbb{E}_{\nu}^{\pi}[T_1^{\pi}(n)] + \Delta_2(\nu)\mathbb{E}_{\nu}^{\pi}[T_2^{\pi}(n)] \\ &= R_n(\pi, \nu). \end{aligned}$$

Therefore,

$$R_n(\pi', \nu) \leq R_n(\pi, \nu)$$

for every $\nu \in \mathcal{E}$.

(b) If $\mu_2 \geq 1$ or $\mu_2 \leq 0$, then (b) is false(Consider trivial policy, choosing optinal arm all the time). Let's suppose $\mu_2 \in (0, 1)$. Then

$$\exists\, p, q \in (0, 1) \text{ such that } 0 < p < \mu_2 < q < 1.$$

Define $\nu_p = (\mathcal{B}(p), \delta_{\mu_2})$, $\nu_q = (\mathcal{B}(q), \delta_{\mu_2})$ and $\tau = \inf\{t \in \mathbb{N} : A_t = 2\}$, also let a retirement policy π be given. Then $T_1(n) = \min\{\tau - 1, n\}$, and $T_2(n) = n - T_1(n)$. If

$$\limsup_{n\to\infty} \frac{R_n(\pi, \nu_p)}{n} > 0,$$

then the conclusion follows. Suppose

$$\limsup_{n\to\infty} \frac{R_n(\pi, \nu_p)}{n} = 0.$$

Then

$$\limsup_{n\to\infty} \frac{(\mu_2 - p)\mathbb{E}_{\nu_p,\pi}[T_1(n)]}{n} = 0.$$

But $T_1(n)/n = \min\{\tau - 1, n\}/n \to \mathbb{1}_{\{\tau=\infty\}}$ almost surely. Since $0 \leq T_1(n)/n \leq 1$, by the dominated convergence theorem,

$$\lim_{n\to\infty} \frac{\mathbb{E}_{\nu_p,\pi}[T_1(n)]}{n} = \mathbb{P}_{\nu_p,\pi}[\tau = \infty] = 0.$$

Therefore,

$$\begin{aligned} &\mathbb{P}_{\nu_q,\pi}[\tau = \infty] = 1 \\ &\implies \mathbb{P}_{\nu_q,\pi}\left[\bigcup_{m=1}^{\infty} \{\tau \leq m\}\right] = 1 \\ &\implies \exists\, m \in \mathbb{N} \text{ such that } \mathbb{P}_{\nu_q,\pi}[\tau \leq m] > 0. \end{aligned}$$

Here $\mathbb{P}_{\nu_p} << \mathbb{P}_{\nu_q}$ since $^\forall y = (y_1, y_2, \ldots) \in \{0,1\}^m$,

$$\begin{aligned}&\mathbb{P}_{\nu_p}[Y_1 = y_1, \ldots, Y_m = y_m] = p^{\sum_{i=1}^m y_i}(1-p)^{m-\sum_{i=1}^m y_i} > 0\\ &\implies \mathbb{P}_{\nu_q}[Y_1 = y_1, \ldots, Y_m = y_m] = q^{\sum_{i=1}^m y_i}(1-q)^{m-\sum_{i=1}^m y_i} > 0.\end{aligned}$$

Hence if $\mathbb{P}_{\nu_q}[\tau \le m] > 0$, then $\mathbb{P}_{\nu_p}[\tau \le m] > 0$. Now let $c = \mathbb{P}_{\nu_q}[\tau \le m] > 0$. Then we have

$$\Delta_2(\nu_q) = q - \mu_2 > 0, \quad \text{and} \quad T_2(n) \ge (n-m+1)\mathbb{1}_{\{\tau \le m\}}.$$

Therefore

$$\begin{aligned}\limsup_{n\to\infty} \frac{R_n(\pi,\nu_q)}{n} &\ge \liminf_{n\to\infty} \frac{R_n(\pi,\nu_q)}{n}\\ &\ge \liminf_{n\to\infty}(q-\mu_2)\frac{\mathbb{E}_{\nu_q,\pi}[T_2(n)]}{n}\\ &\ge \liminf_{n\to\infty}(q-\mu_2)\frac{\mathbb{E}_{\nu_q,\pi}[(n-m+1)\mathbb{1}_{\{\tau\le m\}}]}{n}\\ &= (q-\mu_2)(1-\frac{m-1}{n})\mathbb{P}_{\nu_q,\pi}[\tau \le m]\\ &= (q-\mu_2)c\\ &> 0.\end{aligned}$$

□

5 Concentration of measure

Definition. Let $X, X_1, X_2, \ldots, X_n$ be i.i.d. random variables with real values, and assume that the mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{V}[X] < \infty$. Then the **empirical mean** is defined by

$$\hat{\mu} := \frac{1}{n}\sum_{i=1}^n X_i.$$

This is an unbiased estimator of the mean μ, since $\mathbb{E}[\hat{\mu}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] = \mu$. Using independence of the X_i's, the variance of the empirical mean is

$$\mathbb{V}[\hat{\mu}] = \mathbb{E}\left[(\hat{\mu}-\mu)^2\right] = \frac{\sigma^2}{n}.$$

Two quantities

$$\mathbb{P}[\hat{\mu} \ge \mu + \epsilon] \quad \text{and} \quad \mathbb{P}[\hat{\mu} \le \mu - \epsilon]$$

are called the **tail probabilities** of $\hat{\mu} - \mu$.

Lemma 5.1. *Let X be a random variable and let $\epsilon > 0$. Then*

(a) (Markov's inequality) : $\mathbb{P}[|X| \ge \epsilon] \le \frac{\mathbb{E}[|X|]}{\epsilon}$.

(b) (Chebyshev's inequality) : $\mathbb{P}[|X-\mu| \ge \epsilon] \le \frac{\sigma^2}{\epsilon^2}$.

Proof. (a)

$$\begin{aligned}\mathbb{E}[|X|] &= \int_0^\infty \mathbb{P}[|X| \ge t]\,dt\\ &\ge \int_0^\epsilon \mathbb{P}[|X| \ge \epsilon]\,dt\\ &= \epsilon\mathbb{P}[|X| \ge \epsilon].\end{aligned}$$

(b)

$$\begin{aligned}\mathbb{P}[|X-\mu| \geq \epsilon] &= \mathbb{P}[(X-\mu)^2 \geq \epsilon^2] \\ &\leq \frac{\mathbb{E}[(X-\mu)^2]}{\epsilon^2}.\end{aligned}$$

□

Theorem (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with finite second moment. Define*

$$S_n := \sum_{t=1}^{n} (X_t - \mu).$$

Then for $\sigma^2 > 0$,

$$\frac{S_n}{\sqrt{n}\sigma} \xrightarrow{d} N(0,1).$$

Definition 5.2. A random variable X is **σ-subgaussian** if

$$^{\forall}\lambda \in \mathbb{R}, \quad \mathbb{E}\left[e^{\lambda X}\right] \leq e^{\frac{\lambda^2\sigma^2}{2}}.$$

The **moment generating function** of X is a function

$$M_X(\lambda) := \mathbb{E}\left[e^{\lambda X}\right], \quad \lambda \in \mathbb{R}.$$

The **cumulant generating function** of X is a function

$$\Lambda_X(\lambda) := \log M_X(\lambda) = \log \mathbb{E}\left[e^{\lambda X}\right], \quad \lambda \in \mathbb{R}.$$

Remark. X is σ-subgaussian if and only if

$$\Lambda_X(\lambda) \leq \frac{\lambda^2\sigma^2}{2} \quad \forall \lambda \in \mathbb{R}.$$

Theorem 5.3. *If X is σ-subgaussian, then for all $\epsilon \geq 0$,*

$$\mathbb{P}[X \geq \epsilon] \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Proof. Let $\lambda > 0$. By Markov's inequality,

$$\mathbb{P}[X \geq \epsilon] = \mathbb{P}[e^{\lambda X} \geq e^{\lambda\epsilon}] \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda\epsilon}} \leq \frac{e^{\frac{\lambda^2\sigma^2}{2}}}{e^{\lambda\epsilon}} = \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda\epsilon\right).$$

Letting $\lambda = \frac{\epsilon}{\sigma^2}$ gives

$$\mathbb{P}[X \geq \epsilon] \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

□

Lemma 5.4. *Suppose X is σ-subgaussian and X_1, X_2 are independent σ_1, σ_2-subgaussian, respectively. Then*

(a) $\mathbb{E}[X] = 0$ and $\mathbb{V}[X] \leq \sigma^2$.

(b) cX is $|c|\sigma$-subgaussian for all $c \in \mathbb{R}$.

(c) $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$-subgaussian.

Proof. $\log x$ is convex on $(0, \infty)$, in the sense that for any $x, y > 0$ and $\theta \in [0, 1]$,

$$\log(\theta x + (1-\theta)y) \geq \theta \log(x) + (1-\theta)\log(y).$$

(a) By Jensen's inequality, $\log e^{\frac{\lambda^2\sigma^2}{2}} \geq \log \mathbb{E}[e^{\lambda X}] \geq \mathbb{E}[\log e^{\lambda X}] = \lambda \mathbb{E}[X]$. That is, $\mathbb{E}[X] \leq \frac{\lambda\sigma^2}{2}$ for all $\lambda \in \mathbb{R}$, which implies $\mathbb{E}[X] = 0$. Also,

$$\mathbb{E}[e^{\lambda X}] + \mathbb{E}[e^{-\lambda X}] \leq 2e^{\frac{\lambda^2\sigma^2}{2}}, \quad {}^\forall\lambda \in \mathbb{R}.$$

Hence,

$$\mathbb{E}\left[\frac{e^{\lambda X} - e^{-\lambda X} - 2}{\lambda^2}\right] \leq \frac{e^{\frac{2(\lambda^2\sigma^2)-1}{2}}}{\lambda^2}.$$

Here

$$\frac{e^{\lambda X} - e^{-\lambda X}}{\lambda^2} = \frac{2(\cosh(\lambda X) - 1)}{\lambda^2},$$

which is non-negative and $\frac{2(\cosh(\lambda X)-1)}{\lambda^2} \to X^2$ as $\lambda \to 0$. Therefore, by Fatou's lemma,

$$\mathbb{E}[X^2] \leq \liminf_{\lambda\to 0} \mathbb{E}\left[\frac{2(\cosh(\lambda X) - 1)}{\lambda^2}\right] \leq \lim_{\lambda\to 0} \frac{e^{\frac{2(\lambda^2\sigma^2)-1}{2}}}{\lambda^2} = \sigma^2.$$

(b) ${}^\forall\lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda cX}] = \mathbb{E}[e^{(c\lambda)X}] \leq e^{\frac{(c\lambda)^2\sigma^2}{2}} = e^{\frac{\lambda^2(c\sigma)^2}{2}}$, which shows that cX is $|c|\sigma$-subgaussian.

(c) ${}^\forall\lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda(X_1+X_2)}] = \mathbb{E}[e^{\lambda X_1}]\mathbb{E}[e^{\lambda X_2}] \leq e^{\frac{\lambda^2\sigma_1^2}{2}} e^{\frac{\lambda^2\sigma_2^2}{2}} = e^{\frac{\lambda^2(\sigma_1^2+\sigma_2^2)}{2}}$, which shows that $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$-subgaussian.

□

Corollary 5.5. *Assume $X_i - \mu$ are independent, σ-subgaussian random variables. Then for any $\epsilon \geq 0$,*

$$\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) \quad \textit{and} \quad \mathbb{P}(\hat{\mu} \leq \mu - \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right),$$

where $\hat{\mu} = \frac{1}{n}\sum_{i=1}^n X_i$.

Proof. By Lemma 5.4, $\hat{\mu} - \mu$ is $\frac{\sigma}{\sqrt{n}}$-subgaussian. Therefore, for any $\epsilon \geq 0$,

$$\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) \quad \text{and} \quad \mathbb{P}(\hat{\mu} \leq \mu - \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right)$$

by Theorem 5.3. □

Exercise 5.6.

Exercise 5.9.

Exercise 5.10.

6 The Explore-Then-Commit Algorithm

From now on, we assume that all bandit instances are in $\mathcal{E}_{\text{SG}}^k(1)$, which means the reward distribution of all arms is 1-subgaussian. Let $\hat{\mu}_i(t)$ be the average reward received from arm i up to round t. Formally,

$$\hat{\mu}_i(t) = \frac{1}{T_i(t)} \sum_{s=1}^t \mathbb{1}\{A_s = i\} X_s,$$

where $T_i(t) = \sum_{s=1}^t \mathbb{1}\{A_s = i\}$ is the number of times arm i has been played up to round t.

Input: m

1 In round t choose action

$$A_t = \begin{cases} (t \bmod k) + 1, & \text{if } t \leq mk, \\ \operatorname{argmax}_i \hat{\mu}_i(mk) & \text{if } t > mk, \end{cases}$$

Algorithm 1: ETC Algorithm

Theorem 6.1. *Suppose ETC with 1-subgaussian bandit and* $1 \le m \le n/k$. *Then*

$$R_n \le m \sum_{i=1}^{k} \Delta_i + (n - mk) \sum_{i=1}^{k} \Delta_i \exp\left(-\frac{m\Delta_i^2}{4}\right)$$

Proof. Fill. □

Exercise 6.6 (Doubling trick)**.** Let a fixed sequence $\{n_i\}_{i=1}^{\infty}$ be given and the policy π be given with following regret bound;

$$\max_{1 \le t \le n} R_t(\pi(n), \nu) \le f_n(\nu), \quad {}^{\forall}\nu \in \mathcal{E}.$$

Consider the policy π^* that runs π until round $t = \min\{n, n_1\}$, then restarts π until round $t = \min\{n, n_1 + n_2\}$, and so on.

(a) Let $n > 0$ be arbitrary, $l_{\max} = \min\{l : \sum_{i=1}^{l} n_i \ge n\}$. Prove that ${}^{\forall}\nu \in \mathcal{E}$,

$$R_n(\pi^*, \nu) \le \sum_{l=1}^{l_{\max}} f_{n_l}(\nu)$$

(b) Suppose that $f_n(\nu) \le \sqrt{n}$. Prove that if $n_l = 2^{l-1}$, then for any $\nu \in \mathcal{E}$ and horizon n,

$$R_n(\pi^*, \nu) \le C\sqrt{n}.$$

(c) Suppose that $f_n(\nu) = g(\nu)\log(n)$ for some function $g : \mathcal{E} \to [0, \infty)$. What is the regret of π^* if $n_l = 2^{l-1}$?

Solution 6.6.

(a) *Proof.* Let $N_0 = 0$, $N_l = \sum_{i=1}^{l} n_i$. Then $N_{l_{\max}} < n \le N_{l_{\max}} + n_{l_{\max}}$. Also define last period $m_{l_{\max}} = n - N_{l_{\max}-1}$. Then

$$R_n(\pi^*, \nu) = \sum_{l=1}^{l_{\max}-1} R_{n_l}(\pi(n_l), \nu) + R_{n_{l_{\max}}}(\pi(m_{l_{\max}}), \nu) \le \sum_{l=1}^{l_{\max}-1} f_{n_l}(\nu) + f_{n_{l_{\max}}}(\nu) \le \sum_{l=1}^{l_{\max}} f_{n_l}(\nu).$$

□

(b) *Proof.* Using (a),

$$R_n(\pi^*, \nu) \le \sum_{l=1}^{l_{\max}} f_{n_l}(\nu) \le \sum_{l=1}^{l_{\max}} \sqrt{n_l} \le \sum_{l=1}^{l_{\max}} 2^{\frac{l-1}{2}}.$$

Also, $n > \sum_{l=1}^{l_{\max}-1} n_l = \sum_{l=1}^{l_{\max}-1} 2^{l-1} = 2^{l_{\max}-1} - 1$, so $n \ge 2^{l_{\max}-1} = n_{l_{\max}}$. Therefore,

$$\sum_{l=1}^{l_{\max}} 2^{\frac{l-1}{2}} \le \sum_{j=0}^{l_{\max}} 2^{\frac{j}{2}} = \sum_{j=0}^{l_{\max}} 2^{\frac{l_{\max}-1}{2}} \cdot 2^{-\frac{l_{\max}-1-j}{2}} = \sqrt{2^{l_{\max}-1}} \sum_{j=0}^{l_{\max}} \sqrt{2}^{1-j} \le \sqrt{n} \sum_{j=0}^{\infty} \sqrt{2}^{1-j}.$$

□

(c) *Proof.* Using (a),

$$R_n(\pi^*, \nu) \le \sum_{l=1}^{l_{\max}} g(\nu)\log(2^{l-1}) = g(\nu)\log(2) \sum_{l=1}^{l_{\max}} (l-1) = g(\nu)\log(2)\frac{(l_{\max}-1)l_{\max}}{2}.$$

Here $\sum_{l=1}^{l_{\max}} 2^{l-1} \ge n$, which implies $l_{\max} \ge \log_2(n+1)$. That is, $l_{\max} = \lceil \log_2(n+1) \rceil$. Therefore,

$$R_n(\pi^*, \nu) \le g(\nu)\log(2)\frac{(\lceil \log_2(n+1) \rceil - 1)\lceil \log_2(n+1) \rceil}{2} = Cg(\nu)\mathcal{O}((\log_2(n))^2).$$

□

Exercise 6.8. Assume that arm 1 is an optimal arm and the horizon n is known. Consider following algorithm:

Input: k and sequence $(m_l)_l$
$A_1 = \{1, 2, \ldots, k\}$
for $l = 1, 2, 3, \ldots$ **do**
    Choose each arm $i \in A_l$ exactly m_l times
    Let $\hat{\mu}_{i,l}$ be the average reward for arm i from this phase only
    Update active set:

$$A_{l+1} = \left\{ i : \hat{\mu}_{i,l} + 2^{-l} \geq \max_{j \in A_l} \hat{\mu}_{j,l} \right\}$$

end for

Algorithm 2: Phased elimination for finite-armed bandits

(a) Show that for any $\ell \geq 1$,

$$\mathbb{P}(1 \notin A_{\ell+1},\, 1 \in A_\ell) \leq k \exp\left(-\frac{m_\ell 2^{-2\ell}}{4} \right).$$

(b) Show that if $i \in [k]$ and $\ell \geq 1$ are such that $\Delta_i \geq 2^{-\ell}$, then

$$\mathbb{P}(i \in A_{\ell+1},\, 1 \in A_\ell,\, i \in A_\ell) \leq \exp\left(-\frac{m_\ell (\Delta_i - 2^{-\ell})^2}{4} \right).$$

(c) Let $\ell_i = \min\{\ell \geq 1 : 2^{-\ell} \leq \Delta_i/2\}$. Choose m_ℓ in such a way that $\mathbb{P}(\exists\, \ell : 1 \notin A_\ell) \leq 1/n$ and $\mathbb{P}(i \in A_{\ell_i+1}) \leq 1/n$.

(d) Show that your algorithm has regret at most

$$R_n \leq C \sum_{i:\Delta_i > 0} \left(\Delta_i + \frac{1}{\Delta_i} \log(n) \right),$$

where $C > 0$ is a carefully chosen universal constant.

(e) Modify your choice of m_ℓ and show that the regret of the resulting algorithm satisfies

$$R_n \leq C \sum_{i:\Delta_i > 0} \left(\Delta_i + \frac{1}{\Delta_i} \log \max\{e, n\Delta_i^2\} \right).$$

(f) Show that with an appropriate universal constant $C' > 0$, the regret satisfies

$$R_n \leq \sum_i \Delta_i + C' \sqrt{nk \log(k)}.$$

Solution. Let i'th arm gets reward sample $X_{i,1}, \ldots, X_{i,m_l}$.

(a) *Proof.* Let $l \in \{1, \ldots, n-1\}$. Then

$$\begin{aligned} \mathbb{P}[1 \notin A_{l+1}, 1 \in A_l] &= \mathbb{P}[1 \in A_l, \exists\, i \in A_l \text{ s.t. } \hat{\mu}_{1,l} + 2^{-l} < \hat{\mu}_{i,l}] \\ &\leq \sum_{i=1}^{k} \mathbb{P}[\hat{\mu}_{1,l} + 2^{-l} < \hat{\mu}_{i,l}]. \end{aligned}$$

Here $\hat{\mu}_{i,l} = \frac{1}{m_l}\sum_{j=1}^{m_l} X_{i,j}$. It follows that $^\forall\lambda \in \mathbb{R}$,

$$\begin{aligned}
\mathbb{E}[e^{\lambda(\hat{\mu}_{i,l}-\mu_i)}] &= \mathbb{E}[e^{\frac{\lambda}{m_l}\sum_{j=1}^{m_l}(X_{i,j}-\mu_i)}] \\
&= \prod_{j=1}^{m_l} \mathbb{E}[e^{\frac{\lambda}{m_l}(X_{i,j}-\mu_i)}] \quad \text{(independent)} \\
&\le \prod_{j=1}^{m_l} \exp\left(\frac{1}{2}\left(\frac{\lambda}{m_l}\right)^2\right) \quad \text{(1-subgaussian)} \\
&= \exp\left(\frac{\lambda^2}{2m_l}\right)
\end{aligned}$$

Therefore $\hat{\mu}_{i,l} - \mu_i$ is $\frac{1}{m_l}$-subgaussian. Hence $(\hat{\mu}_{i,l} - \hat{\mu}_{1,l}) - (\mu_i - \mu_1)$ is $\frac{2}{m_l}$-subgaussian. It follows that

$$\begin{aligned}
\mathbb{P}[\hat{\mu_{1,l}} + 2^{-l} < \hat{\mu}_{i,l}] &\le \mathbb{P}[(\hat{\mu_{i,l}} - \hat{\mu_{1,l}}) - (\mu_i - \mu_1) > 2^{-l}] \quad (\because \mu_i - \mu_1 < 0) \\
&\le \exp\left(-\frac{2^{-2l}}{2\cdot\frac{2}{m_l}}\right) \quad \text{(Theorem 5.3)} \\
&= \exp\left(-\frac{m_l 2^{-2l}}{4}\right).
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbb{P}[1 \notin A_{l+1}, 1 \in A_l] &\le \sum_{i=1}^{k} \exp\left(-\frac{m_l 2^{-2l}}{4}\right) \\
&= k\exp\left(-\frac{m_l 2^{-2l}}{4}\right).
\end{aligned}$$

□

(b) *Proof.* Let $i \in \{1,\dots,k\}$ and $\ell \ge 1$ be $\Delta_i \ge 2^{-l}$. Then

$$\begin{aligned}
\mathbb{P}[i \in A_{l+1}, 1 \in A_l, i \in A_l] &\le \mathbb{P}[\hat{\mu}_{i,l} + 2^{-l} \ge \hat{\mu}_{1,l}] \\
&= \mathbb{P}[(\hat{\mu}_{1,l} - \hat{\mu}_{1,l}) - (\mu_1 - \mu_i) \le 2^{-l} - \Delta_i] \\
&= \mathbb{P}[-(\hat{\mu}_{1,l} - \hat{\mu}_{i,l}) + (\mu_1 - \mu_i) \ge \Delta_i - 2^{-l}].
\end{aligned}$$

By the similiar argument with (a), $-\left((\hat{\mu}_{1,l} - \hat{\mu}_{i,l}) + (\mu_1 - \mu_i)\right)$ is $\frac{2}{m_l}$-subgaussian.

$$\begin{aligned}
\therefore \mathbb{P}[i \in A_{l+1}, 1 \in A_l, i \in A_l] &\le \exp\left(-\frac{(\Delta_i - 2^{-l})^2}{2\cdot\frac{2}{m_l}}\right) \\
&= \exp\left(-\frac{m_l(\Delta_i - 2^{-l})^2}{4}\right).
\end{aligned}$$

□

(c) *Proof.* Fix $i \in \{1,\dots,k\}$. Then

$$\mathbb{P}\left[\exists \ell_0 \text{ s.t. } 1 \notin A_{\ell_0}\right] = \mathbb{P}\left[\bigcup_{\ell\ge 1}\{1 \in A_\ell, 1 \notin A_{\ell+1}\}\right] \le \sum_{\ell\ge 1} k\exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right)$$

by (a). Choose $m_\ell = 2^{2\ell+2}\log(2^{\ell+1}\cdot nk)$. Then

$$\sum_{\ell\ge 1} k\exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right) = \sum_{\ell\ge 1} k\cdot 2^{-\ell-1}\cdot\frac{1}{nk} = \frac{1}{2n}.$$

Also, $\{i \in A_{i+1}\} \subset \{i \in A_{\ell_i+1}, 1 \in A_{\ell_i}, i \in A_{\ell_i}\} \cup \{1 \notin A_{\ell_i}\}$ implies

$$\begin{aligned}
\mathbb{P}\left[i \in A_{i+1}\right] &\le \mathbb{P}\left[i \in A_{\ell_i+1}, 1 \in A_{\ell_i}, i \in A_{\ell_i}\right] + \mathbb{P}\left[1 \notin A_{\ell_i}\right] \\
&\le \exp\left(-\frac{m_{\ell_i}\left(\Delta_i - 2^{-2\ell_i}\right)^2}{4}\right) + \mathbb{P}\left[\exists \ell \text{ s.t. } 1 \notin A_\ell\right] \\
&\le \exp\left(-\frac{m_{\ell_i} \cdot \frac{\Delta_i^2}{4}}{4}\right) + \mathbb{P}\left[\exists \ell \text{ s.t. } 1 \notin A_\ell\right] \\
&\le \exp\left(-\frac{2^{2\ell_i} \cdot \Delta_i^2 \cdot \log(2^{\ell_i+1} \cdot nk)}{4}\right) + \frac{1}{2n} \\
&\le \exp\left(-\log(2^{\ell_i+1} \cdot nk)\right) + \frac{1}{2n} \\
&\le 2^{-\ell_i-1} \cdot \frac{1}{nk} + \frac{1}{2n} \\
&\le \frac{1}{n}
\end{aligned}$$

for $m_\ell = \lceil 2^{2\ell+2} \log(2^{\ell+1} \cdot nk) \rceil$. □

(d) Suppose $\max_i \Delta_i \le 1$ and $n \ge 2$. Let

$$m_\ell = \lceil 2^{2\ell+2} \log(4n^2) \rceil \qquad \text{and} \qquad \tau_\ell = \sum_{s=1}^{\ell} |A_s| m_s$$

Also define

$$G = \bigcap_{\ell:\tau_\ell \le n} \bigcap_{i \in A_\ell \setminus \{1\}} \left(\{1 \notin A_\ell\} \cup \left\{|(\hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell}) - (\mu_i - \mu_1)| < 2^{-\ell}\right\}\right).$$

Then, by definition of m_ℓ,

$$2\exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right) \le \frac{1}{2n^2},$$

Also note that subgaussian tail bound implies

$$\mathbb{P}[|(\hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell}) - (\mu_i - \mu_1)| \ge 2^{-\ell}] \le 2\exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right).$$

Hence we get

$$\begin{aligned}
\mathbb{P}[G^c] &= \mathbb{P}\left[\bigcup_{\ell:\tau_\ell \le n} \bigcup_{i \in A_\ell \setminus \{1\}} \left(\{1 \in A_\ell\} \cap \left\{|(\hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell}) - (\mu_i - \mu_1)| \ge 2^{-\ell}\right\}\right)\right] \\
&\le \sum_{\ell \ge 1} \sum_{i \ne 1} \mathbb{E}\left[\mathbb{1}_{\{\tau_\ell \le n\}} \mathbb{1}_{\{1, i \in A_\ell\}} \mathbb{1}_{\{|(\hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell}) - (\mu_i - \mu_1)| \ge 2^{-\ell}\}}\right] \\
&\le \frac{1}{2n^2} \sum_{\ell \ge 1} \sum_{i \ne 1} \mathbb{E}\left[\mathbb{1}_{\{\tau_\ell \le n\}} \mathbb{1}_{\{1, i \in A_\ell\}}\right] \\
&= \frac{1}{2n^2} \sum_{\ell \ge 1} \sum_{i \ne 1} \mathbb{E}\left[\mathbb{1}_{\{\tau_\ell \le n\}} \mathbb{1}_{\{1 \in A_\ell\}} \mathbb{1}_{\{i \in A_\ell\}}\right] \\
&\le \frac{1}{2n^2} \sum_{\ell \ge 1} \mathbb{E}\left[\mathbb{1}_{\{\tau_\ell \le n\}} m_\ell |A_\ell|\right] \\
&\le \frac{1}{2n^2} n = \frac{1}{2n}.
\end{aligned}$$

If G happens, arm-1 is not eliminated for all phases until n and suboptimal arm arm-i is eliminated if phase ℓ_i is successfully completed unitl n where $i \in A_{\ell_i}$. Even if the phase ℓ_i were not successfully completed and stopped at n, $T_i(n) \leq \sum_{\ell}^{\ell_i} m_\ell$. Hence

$$T_i(n) \leq \sum_{\ell=1}^{\ell_i} m_\ell + n\mathbb{1}_{G^c},$$

which implies

$$\mathbb{E}[T_i(n)] \leq \sum_{\ell=1}^{\ell_i} m_\ell + \frac{1}{2}$$

7 The Upper Confidence Bound Algorithm

Note that for sequence of independent 1-subgaussian random variables $(X_t)_{t=1}^n$, mean μ and the empirical mean $\hat{\mu}_n = \frac{1}{n}\sum_{t=1}^n X_t$ satisfy

$$\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\log(1/\delta)}{n}}\right) \leq \delta, \quad {}^\forall\delta \in (0,1).$$

Define

$$\mathrm{UCB}_i(t-1,\delta) = \begin{cases} \infty, & \text{if } T_i(t-1) = 0, \\ \hat{\mu}_i(t-1) + \sqrt{\frac{2\log(1/\delta)}{T_i(t-1)}}, & \text{if } T_i(t-1) > 0. \end{cases}$$

Input k and δ
for $t \in \{1,\ldots,n\}$ **do**
    Choose action

$$A_t = \operatorname*{argmax}_i \mathrm{UCB}_i(t-1,\delta)$$

    Observe reward X_t and update upper confidence bounds
end for

Algorithm 3: UCB(δ)

Theorem 7.1. *Suppose UCB with stochastic k-armed 1-subgaussian bandits. Then for all n, if $\delta = 1/n$, then*

$$R_n \leq 3\sum_{i=1}^k \Delta_i + \sum_{i:\Delta_i>0} \frac{16\log(n)}{\Delta_i}$$

Proof. WLOG, assume that $\mu_1 = \mu^*$. Let G_i be the 'good' event defined by

$$G_i = \left\{\mu_1 < \min_{t\in\{1,\ldots,n\}} \mathrm{UCB}_1(t,\delta)\right\} \cap \left\{\hat{\mu}_{i,u_i} + \sqrt{\frac{2}{u_i}\log\left(\frac{1}{\delta}\right)} < \mu_1\right\}$$

where $u_i \in \{1,\ldots,n\}$. Assume that G_i holds for all $i \in \{1,\ldots,k\}$. To make contradiction, suppose that $T_i(n) > u_i$. Then there exists $t \in \{1,\ldots,n\}$ such that $T_i(t-1) = u_i$ and $A_t = i$. Hence

$$\begin{aligned} \mathrm{UCB}_i(t-1,\delta) &= \hat{\mu}_i(t-1) + \sqrt{\frac{2\log(1/\delta)}{T_i(t-1)}} \\ &= \hat{\mu}_{i,u_i} + \sqrt{\frac{2}{u_i}\log\left(\frac{1}{\delta}\right)} \\ &< \mu_1 \\ &\leq \mathrm{UCB}_1(t-1,\delta). \end{aligned}$$

Therefore, $A_t = \text{argmax}_j \, \text{UCB}_j(t-1,\delta) \neq i$, which is a contradiction. On the other hand,

$$G_i^c = \left\{ \mu_1 \geq \min_{t \in \{1,\dots,n\}} \text{UCB}_1(t,\delta) \right\} \cup \left\{ \hat{\mu}_{i,u_i} + \sqrt{\frac{2}{u_i} \log\left(\frac{1}{\delta}\right)} \geq \mu_1 \right\}.$$

Former part bounded by

$$\begin{aligned}
\mathbb{P}\left(\mu_1 \geq \min_{t \in \{1,\dots,n\}} \text{UCB}_1(t,\delta) \right) &\leq \mathbb{P}\left(\mu_1 \geq \min_{s \in \{1,\dots,n\}} \hat{\mu}_1(s) + \sqrt{\frac{2}{s} \log\left(\frac{1}{\delta}\right)} \right) \\
&\leq \mathbb{P}\left(\bigcup_{s \in \{1,\dots,n\}} \left\{ \mu_1 \geq \hat{\mu}_1(s) + \sqrt{\frac{2}{s} \log\left(\frac{1}{\delta}\right)} \right\} \right) \\
&\leq \sum_{s \in \{1,\dots,n\}} \mathbb{P}\left(\mu_1 \geq \hat{\mu}_1(s) + \sqrt{\frac{2}{s} \log\left(\frac{1}{\delta}\right)} \right) \\
&\leq \sum_{s \in \{1,\dots,n\}} \delta = n\delta.
\end{aligned}$$

For latter part, assume u_i is chosen large enough that

$$\Delta_i - \sqrt{\frac{2}{u_i} \log\left(\frac{1}{\delta}\right)} \geq c\Delta_i$$

for some $c \in (0,1)$. Then

$$\begin{aligned}
\mathbb{P}\left(\hat{\mu}_{i,u_i} + \sqrt{\frac{2}{u_i} \log\left(\frac{1}{\delta}\right)} \geq \mu_1 \right) &= \mathbb{P}\left(\hat{\mu}_{i,u_i} - \mu_i \geq \Delta_i - \sqrt{\frac{2}{u_i} \log\left(\frac{1}{\delta}\right)} \right) \\
&\leq \mathbb{P}\left(\hat{\mu}_{i,u_i} - \mu_i \geq c\Delta_i \right) \\
&\leq \exp\left(-\frac{u_i (c\Delta_i)^2}{2} \right)
\end{aligned}$$

by Cor.5.5. Now we have

$$\mathbb{P}(G_i^c) \leq n\delta + \exp\left(-\frac{u_i (c\Delta_i)^2}{2} \right).$$

Also

$$\mathbb{E}[T_i(n)] = \mathbb{E}[\mathbb{1}_{G_i} T_i(n)] + \mathbb{E}[\mathbb{1}_{G_i^c} T_i(n)].$$

Therefore, choosing

$$u_i = \left\lceil \frac{2\log(1/\delta)}{((1-c)\Delta_i)^2} \right\rceil$$

follows that

$$\mathbb{E}[T_i(n)] \leq u_i + 1 + n^{1-2c^2/(1-c)^2} = \left\lceil \frac{2\log\left(n^2\right)}{((1-c)\Delta_i)^2} \right\rceil + 1 + n^{1-2c^2/(1-c)^2}.$$

Lastly, choosing $c = 1/2$ follows that

$$\mathbb{E}[T_i(n)] \leq \left\lceil \frac{16\log(n)}{\Delta_i^2} \right\rceil + 2,$$

and regret decomposition lemma leads to

$$\begin{aligned} R(n) &\leq \sum_{i:\Delta_i>0} \Delta_i \left(\left\lceil \frac{16\log(n)}{\Delta_i^2} \right\rceil + 2 \right) \\ &= \sum_{i:\Delta_i>0} \left(\frac{16\log(n)}{\Delta_i} + \Delta_i \right) + 2 \sum_{i:\Delta_i>0} \Delta_i \\ &\leq 3\sum_{i=1}^{k} \Delta_i + \sum_{i:\Delta_i>0} \frac{16\log(n)}{\Delta_i}. \end{aligned}$$

□

Theorem 7.2. *Suppose $\nu \in \mathcal{E}_{\mathrm{SG}}^k(1)$ and $\delta = 1/n^2$. Then regret of the UCB algorithm is bounded by*

$$R(n) \leq 8\sqrt{nk\log(n)} + 3\sum_{i=1}^{k} \Delta_i.$$

Proof. Let $\Delta > 0$ be some value to be determined later. Recall from proof of 7.1,

$$\mathbb{E}[T_i(n)] \leq 3 + \frac{16\log(n)}{\Delta_i^2}.$$

Hence regret decomposition lemma follows that

$$\begin{aligned} R_n &= \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(n)] \\ &= \sum_{i:\Delta_i<\Delta} \Delta_i \mathbb{E}[T_i(n)] + \sum_{i:\Delta_i\geq\Delta} \Delta_i \mathbb{E}[T_i(n)] \\ &\leq n\Delta + \sum_{i:\Delta_i\geq\Delta} \left(3\Delta_i + \frac{16\log(n)}{\Delta_i} \right) \\ &\leq n\Delta + 3\sum_{i=1}^{k} \Delta_i + \frac{16k\log(n)}{\Delta} \\ &\leq 8\sqrt{nk\log(n)} + 3\sum_{i=1}^{k} \Delta_i. \end{aligned}$$

by choosing $\Delta = \sqrt{\frac{16k\log(n)}{n}}$. □

Exercise 7.4 (Phased UCB 1)**.** Consider the following algorithm, called phased UCB;

```
Input k and δ
Choose each arm once
for l = 1, 2, ... do
    Compute A_l = argmax_i UCB_i(t − 1, δ)
    Choose arm A_l and exactly 2^l times
end for
```

Algorithm 4: A phased version of UCB

(a) State and prove a bound on the regret for this version of UCB.

(b) Compare with Theorem 7.1.

(c) How would the result change if the lth phase had a length of $\lceil \alpha^l \rceil$ with $\alpha > 1$?

Solution.

(a) Let $k = 2$, L be some large constant(≥ 3), $n = 2^{L+1}$, $\delta = n^{-2}$, and $q = 2\sqrt{\log(n)}$. Also let two arms attain deterministic reward, $\mu_1 = 0$, $\mu_2 = -\Delta$, respectively. ($\Delta > 0$) In this setting, the algorithm chooses each arm once initially and then goes to round 3. Here learner gets UCB

$$UCB_1(2, \delta) = q, \quad UCB_2(2, \delta) = -\Delta + q$$

and hence chooses arm 1 in round 3, and 4. Explicit simulation will be following;

t	l	UCB_1	UCB_2	Next action
1	0	∞	∞	1
2	0	q	∞	2
3	1	q	$-\Delta + q$	1
4	1	$q/\sqrt{2}$	$-\Delta + q$	1
5	2	$q/\sqrt{3}$	$-\Delta + q$	1
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
2^L	$L-1$	$q/\sqrt{2^L - 2}$	$-\Delta + q$	1
$2^L + 1$	L	$q/\sqrt{2^L - 1}$	$-\Delta + q$	2
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

if we choose Δ with

$$\frac{q}{\sqrt{2^L - 1}} < q - \Delta < \frac{q}{\sqrt{2^{L-1} - 1}}.$$

Then the regret will be dominated by the suboptimal arm 2 in the last phase, giving

$$R_n \geq \Delta \cdot 2^L = \frac{n}{2}\Delta.$$

This example shows that the phased UCB algorithm can obtain linear regret,

$$R_n \leq n \cdot \max_i \Delta_i,$$

which is trivial.

(b) The above example of (a) shows that the phased UCB algorithm can get linear regret, while Theorem 7.1 shows that the standard UCB always get lower than sublinear regret.

(c) Still get linear regret for the phased UCB algorithm with any constant $\alpha > 1$.

Exercise 7.5 (Phased UCB 2)**.** Consider the following another phased version of UCB;

Input k and δ
Choose each arm once
for $l = 1, 2, \ldots$ **do**
    Let $t_l = t$
    Compute $A_l = \text{argmax}_i \, \text{UCB}_i(t_l - 1, \delta)$
    Choose arm A_l until round t such that $T_{A_l}(t) \geq \alpha T_{A_l}(t_l - 1)$
end for

(a) State and prove a bound on the regret for this version of UCB with $\alpha = 2$.

(b) Compare with previous exercise and Theorem 7.1.

(c) Repeat (a) and (b) for $\alpha > 1$.

(d) Implement theses alogorithms and compare them empirically to UCB(δ)

Solution. WLOG, assume that arm 1 is optimal.

(a) For all suboptimal arms i, let

$$H_i = \left\{\forall t \in \{1,\dots,n\}, \mu_1 < \hat{\mu}_{1,t} + \sqrt{\frac{2\log(1/\delta)}{t}}\right\} \cap \left\{\forall t \in \{1,\dots,n\}, \mu_i > \hat{\mu}_{i,t} - \sqrt{\frac{2\log(1/\delta)}{t}}\right\}.$$

Then the complement event H_i^c is

$$H_i^c = \left\{\exists t \in \{1,\dots,n\} : \mu_1 \geq \hat{\mu}_{1,t} + \sqrt{\frac{2\log(1/\delta)}{t}}\right\} \cup \left\{\exists t \in \{1,\dots,n\} : \mu_i \leq \hat{\mu}_{i,t} - \sqrt{\frac{2\log(1/\delta)}{t}}\right\}.$$

That is

$$\mathbb{P}(H_i^c) \leq \sum_{t=1}^{n} \mathbb{P}\left(\mu_1 \geq \hat{\mu}_{1,t} + \sqrt{\frac{2\log(1/\delta)}{t}}\right) + \sum_{t=1}^{n} \mathbb{P}\left(\mu_i \leq \hat{\mu}_{i,t} - \sqrt{\frac{2\log(1/\delta)}{t}}\right).$$

Cor 5.5 follows that

$$\mathbb{P}(H_i^c) \leq 2\sum_{t=1}^{n} \exp\left(-\frac{2\log(1/\delta)}{t} \cdot \frac{t}{2}\right) = 2\sum_{t=1}^{n} \delta = 2n\delta.$$

For the latest phase l where arm i is chosen, let $m := T_i(t_l - 1)$. Upon the event H_i, letting $m \geq u_i$ implies that

$$\mathrm{UCB}_i(t_l - 1, \delta) = \hat{\mu}_{i,m} + \sqrt{\frac{2\log(1/\delta)}{m}} < \mu_i + \sqrt{\frac{8\log(1/\delta)}{m}} < \mu_i + \Delta_i = \mu_1 < \mathrm{UCB}_1(t_l - 1, \delta)$$

where $u_i = \lfloor \frac{8\log(1/\delta)}{\Delta_i^2} \rfloor + 1$.

On the other hand, if $m < u_i$, then $T_i(n) \leq 2m < 2u_i$. Therefore

$$\begin{aligned}\mathbb{E}[T_i(n)] &\leq 2u_i + n\mathbb{P}(H_i^c) \\ &= 2u_i + 2n^2\delta.\end{aligned}$$

Choosing $\delta = n^{-2}$ follows that $u_i \leq \frac{16\log(n)}{\Delta_i^2} + 1$, and hence

$$\mathbb{E}[T_i(n)] \leq 2\left(\frac{16\log(n)}{\Delta_i^2} + 1\right) + 2 = \frac{32\log(n)}{\Delta_i^2} + 4.$$

The regret is bounded by

$$\begin{aligned}R(n) &= \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(n)] \\ &\leq \sum_{i:\Delta_i>0} \Delta_i \left(\frac{32\log(n)}{\Delta_i^2} + 4\right) \\ &= 4\sum_{i=1}^{k} \Delta_i + 32 \sum_{i:\Delta_i>0} \frac{\log(n)}{\Delta_i}.\end{aligned}$$

(b) Slightly larger than UCB, they are differ by scalar factor.

(c) $T_i(n) \leq \lceil \alpha m \rceil$,

$$R(n) \lesssim 4\sum_{i=1}^{k} \Delta_i + 16\alpha \sum_{i:\Delta_i>0} \frac{\log(n)}{\Delta_i}.$$

(d) Refer "./Implementations".

8 The Upper Confidence Bound Algorithm: Asymptotic Optimality

Input k
Choose each arm once
Subsequently choose

$$A_t = \operatorname*{argmax}_i \left(\hat{\mu}_i(t-1) + \sqrt{\frac{2 \log f(t)}{T_i(t-1)}} \right)$$

where $f(t) = 1 + t \log^2(t)$.

Algorithm 5: Asymptotically optimal UCB

Theorem 8.1. *For any 1-subgaussian bandit, regret of the asymptotically optimal UCB policy satisfies*

$$R_n \leq \sum_{i:\Delta_i > 0} \inf_{\epsilon \in (0, \Delta_i)} \Delta_i \left(1 + \frac{5}{\epsilon^2} + \frac{2\left(\log f(n) + \sqrt{\pi \log f(n)} + 1 \right)}{(\Delta_i - \epsilon)^2} \right).$$

Furthermore,

$$\limsup_{n \to \infty} \frac{R_n}{\log n} \leq \sum_{i:\Delta_i > 0} \frac{2}{\Delta_i}.$$

Choosing $\epsilon = \Delta_i/2$ inside the sum shows that

$$R_n \leq \sum_{i:\Delta_i > 0} \left(\Delta_i + \frac{1}{\Delta_i} \left(8 \log f(n) + 8\sqrt{\pi \log f(n)} + 28 \right) \right).$$

That is, $\exists\, C > 0$ such that

$$R_n \leq \sum_{i:\Delta_i > 0} C \left(\Delta_i + \frac{\log n}{\Delta_i} \right).$$

Lemma 8.2. *Let $X_1, \ldots, X_n$ be independent 1-subgaussian random variables, $\hat{\mu}_t = \frac{1}{t} \sum_{i=1}^t X_i, \epsilon > 0, a > 0$. Suppose that*

$$\kappa = \sum_{t=1}^n \mathbb{1}_{\{\hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon\}}, \quad \kappa' = u + \sum_{t=\lceil u \rceil}^n \mathbb{1}_{\{\hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon\}}$$

where $u = 2a\epsilon^{-2}$. Then

$$\mathbb{E}[\kappa] \leq \mathbb{E}[\kappa'] \leq 1 + \frac{2}{\epsilon^2}(a + \sqrt{\pi a} + 1).$$

Exercise 8.2. Consider the one-armed bandit problem: $\mathcal{E} = \{\mathcal{N}(\mu_1, 1) : \mu_1 \in \mathbb{R}\} \times \{\mathcal{N}(0,1)\}$. Suppose that $\nu = (P_1, P_2) \in \mathcal{E}$, and P_1 has mean $\mu_1 = 1$. Evaluate

$$\limsup_{n \to \infty} \frac{R_n(\pi, \nu)}{\log n}$$

where π is the policy of asymptotically optimal UCB.

Exercise 8.3. Consider the setting of Exercise 8.2 and define a policy by

$$A_t = \begin{cases} 1 & \text{if } \hat{\mu_1}(t-1) + \sqrt{\frac{2 \log f(t)}{T_1(t-1)}} \geq 0, \\ 2 & \text{otherwise.} \end{cases}$$

Suppose that $\nu = (P_1, P_2)$ where $P_1 = \mathcal{N}(\mu_1, 1)$ and $P_2 = \mathcal{N}(0,1)$. Prove that for the modified policy,

$$\limsup_{n \to \infty} \frac{R_n(\pi, \nu)}{\log n} \leq \begin{cases} 0 & \text{if } \mu_1 \geq 0, \\ -\frac{2}{\mu_1} & \text{if } \mu_1 < 0. \end{cases}$$

9 The Upper Confidence Bound Algorithm: Minimax Optimality

Here we approach to the minimax optimality with regret bound of worst-case be less than equal to $\sqrt{nk}$, by MOSS(Minimax Optimal Strategy in the Stochastic case) algorithm. But MOSS still requires the horizon n, which we will handle later.

Input n and k
Choose each arm once
Subsequently choose

$$A_t = \operatorname*{argmax}_i \hat{\mu}_i(t-1) + \sqrt{\frac{4}{T_i(t-1)} \log^+ \left(\frac{n}{kT_i(t-1)} \right)},$$

where $\log^+(x) = \log \max\{1, x\}$.

Algorithm 6: MOSS

Theorem 9.1. *For any 1-subgaussian bandit, the regret of MOSS algorithm satisfies*

$$R_n \leq 39\sqrt{kn} + \sum_{i=1}^{k} \Delta_i.$$

Theorem 9.2. *Let $X_1, \ldots, X_n$ be a sequence of independent σ-subgaussian random variables and $S_t = \sum_{i=1}^{t} X_i$. Then $^\forall \epsilon > 0$,*

$$\mathbb{P}[\exists\, t \leq n \text{ s.t. } S_t \geq \epsilon] \leq \exp\left(-\frac{\epsilon^2}{2n\sigma^2} \right).$$

Note. Thm. 9.2 implies Cor. 5.5.

Lemma 9.3. *Let $\delta \in (0,1)$ and $X_1, \ldots$ be independent 1-subgaussian random variables, and $\hat{\mu}_t = \frac{1}{t} \sum_{i=1}^{t} X_i$. Then $^\forall \Delta > 0$,*

$$\mathbb{P}\left[\exists\, s \geq 1 \text{ s.t. } \hat{\mu}_s + \sqrt{\frac{4}{s} \log^+ \left(\frac{1}{s\delta} \right)} + \Delta \leq 0 \right] \leq \frac{15\delta}{\Delta^2}.$$

9.1 Note on minimax optimality of MOSS

In chapter 9, the term "minimax optimality" appears repeatedly, but it has not formally defined. Here we state precisely in what sense MOSS is minimax optimal.

Theorem. *For k, define $\mathcal{E}_{SG}^k(1)$ be the class of all k-armed 1-subgaussian instances. If suboptimal gap $\Delta_i \leq 1$ for all $i \in \{1, \ldots, k\}$, then $\exists\, c, C > 0$ such that $^\forall k \geq 2$ and $n \geq k$,*

$$c\sqrt{kn} \leq \inf_{\pi \in \Pi} \sup_{\nu \in \mathcal{E}_{SG}^k(1)} R_n(\pi, \nu) \leq \sup_{\nu \in \mathcal{E}_{SG}^k(1)} R_n(\pi_{k,n}^{MOSS}, \nu) \leq C\sqrt{kn}$$

where Π is a family of policies for k-armed 1-subgaussian instances with finite horizon n, $\pi_{k,n}^{MOSS}$ is a MOSS algorithm for k-armed 1-subgaussian instances with finite horizon n.

The first inequality is from Theorem 15.2, and the last inequality is from Theorem 9.1. Uniform bound for suboptimal gap is not need to be 1, only constant factor differs if we set arbitrarily. From the theorem, we may say MOSS is minimax optimal up to universal constant factors over the class of k-armed 1-subgaussian instances with known horizon $n \leq k$, where the suboptimal gap is uniformly bounded.

Theorem 15.2 (p.199)**.** *Let $k > 1$ and $n \geq k - 1$. Then, for any policy π, $\exists\, \mu \in [0,1]^k$ such that*

$$R_n(\pi, \nu_\pi) \geq \frac{1}{27}\sqrt{(k-1)n}$$

where ν_π is the Gaussian bandit for which the ith arm has reward distribution $\mathcal{N}(\mu_i, 1)$.

Exercise 9.3 (UCB*)**.** Suppose we modify the index used by MOSS to be

$$\hat{\mu}_i(t-1) + \sqrt{\frac{4}{T_i(t-1)} \log^+\left(\frac{n}{T_i(t-1)}\right)}.$$

(a) Show that for all 1-subgaussian bandits, this new policy suffers regret at most

$$R_n \leq C\left(\sum_{i:\Delta_i>0} \Delta_i + \frac{1}{\Delta_i}\log^+(n\Delta_i^2)\right)$$

where $C > 0$ is a universal constant.

(b) Under the same conditions as the previous part, show there exists a universal constant $C > 0$ such that

$$R_n \leq C\sqrt{kn\log(k)} + \sum_{i=1}^{k} \Delta_i.$$

(c) Repeat part (a) and part (b) using the index

$$\hat{\mu}_i(t-1) + \sqrt{\frac{4}{T_i(t-1)} \log^+\left(\frac{t}{T_i(t-1)}\right)}.$$

Solution 9.3. Suppose $n \geq k \geq 2$.

(a) Define

$$b(s) = \sqrt{\frac{4}{s}\log^+\left(\frac{n}{s}\right)}, \text{ and } U_i(s) = \hat{\mu}_{i,s} + b(s),$$

and an event

$$G_i = \left\{\exists s \in \{1,\dots,n\} \text{ s.t. } U^*(s) < \mu^* - \frac{\Delta_i}{2}\right\}.$$

Apply Lemma 9.3 with $\delta = 1/n$, $\Delta = \Delta_i/2$, Y_j^* be j-th reward of optimal arm, and $X_j = Y_j^* - \mu^*$ implies

$$\begin{aligned}
\mathbb{P}\left[\exists\, s \in \{1,\dots,n\} \text{ s.t. } \frac{1}{s}\sum_{j=1}^{s} X_j + b(s) + \frac{\Delta_i}{2} \leq 0\right] &= \mathbb{P}\left[\exists\, s \text{ s.t. } \hat{\mu}_s^* + b(s) \leq \mu^* - \frac{\Delta_i}{2}\right] \\
&= \mathbb{P}\left[\exists\, s \text{ s.t. } U^*(s) \leq \mu^* - \frac{\Delta_i}{2}\right] \\
&= \mathbb{P}\left[G_i\right] \\
&\leq \frac{60}{n\Delta_i^2}.
\end{aligned}$$

Now, for any suboptimal arm i,

$$\begin{aligned}
T_i(n) &= 1 + \sum_{t=k+1}^{n} \mathbb{1}_{\{A_t=i\}} \\
&= 1 + \sum_{t=k+1}^{n} \mathbb{1}_{\left\{A_t=i,\, U_i(T_i(t-1))\geq \mu_i+\frac{\Delta_i}{2}\right\}} + \mathbb{1}_{\left\{A_t=i,\, U_i(T_i(t-1))<\mu_i+\frac{\Delta_i}{2}\right\}} \\
&\leq 1 + \sum_{s=1}^{T_i(n)-1} \mathbb{1}_{\left\{U_i(s)\geq \mu_i+\frac{\Delta_i}{2}\right\}} + \sum_{t=k+1}^{n} \mathbb{1}_{\left\{U^*(T^*(t-1))\leq U_i(T_i(t-1))<\mu_i+\frac{\Delta_i}{2}\right\}} \\
&\leq 1 + \sum_{s=1}^{n} \mathbb{1}_{\left\{U_i(s)\geq \mu_i+\frac{\Delta_i}{2}\right\}} + \sum_{s=1}^{n} \mathbb{1}_{\left\{U^*(s)<\mu^*-\frac{\Delta_i}{2}\right\}} \\
&\leq 1 + n\mathbb{1}_{G_i} + \sum_{s=1}^{n} \mathbb{1}_{\left\{U_i(s)\geq \mu_i+\frac{\Delta_i}{2}\right\}}.
\end{aligned}$$

Hence

$$\mathbb{E}\left[T_i(n)\right] \leq 1 + n\mathbb{P}\left[G_i\right] + \sum_{s=1}^{n} \mathbb{P}\left[U_i(s) \geq \mu_i + \frac{\Delta_i}{2}\right].$$

Here the summand is equal to $\mathbb{P}\left[\hat{\mu}_{i,s} - \mu_i \geq \frac{\Delta_i}{2} - b(s)\right]$, whose bound will be determined by Cor 5.5. So, we shall found separating value that makes $\frac{\Delta_i}{2} - b(s) > 0$.

$$\begin{aligned}
b(s) \leq \frac{\Delta_i}{4} &\iff s\Delta_i^2 \geq 64\log^+\frac{n}{s} \\
&\impliedby s\Delta_i^2 \geq 64\log^+(n\Delta_i^2) \text{ and } s \geq \frac{1}{\Delta_i^2} \\
&\impliedby s \geq \frac{64\log^+(n\Delta_i^2)}{\Delta_i^2}.
\end{aligned}$$

Separate s by

$$s \leq \left\lceil \frac{64\log^+(n\Delta_i^2)}{\Delta_i^2} \right\rceil - 1, \quad s \geq \left\lceil \frac{64\log^+(n\Delta_i^2)}{\Delta_i^2} \right\rceil.$$

Then

$$\begin{aligned}
\mathbb{E}\left[T_i(n)\right] &\leq 1 + n\mathbb{P}\left[G_i\right] + \left\lceil \frac{64\log^+(n\Delta_i^2)}{\Delta_i^2} \right\rceil - 1 + \sum_{s=\left\lceil \frac{64\log^+(n\Delta_i^2)}{\Delta_i^2} \right\rceil}^{n} \mathbb{P}\left[\hat{\mu_{i,s}} - \mu_i \geq \frac{\Delta_i}{4}\right] \\
&\leq \frac{60}{\Delta_i^2} + \left\lceil \frac{64\log^+(n\Delta_i^2)}{\Delta_i^2} \right\rceil + \sum_{s=1}^{\infty} \exp\left(-\frac{s\Delta_i^2}{32}\right) \\
&\leq 1 + \frac{60 + 64\log^+(n\Delta_i^2)}{\Delta_i^2} + \frac{32}{\Delta_i^2} \\
&= 1 + \frac{92 + 64\log^+(n\Delta_i^2)}{\Delta_i^2}.
\end{aligned}$$

$$\therefore R_n = \sum_{i=1}^{n} \Delta_i \mathbb{E}[T_i(n)] \leq C\left(\sum_{i=1}^{n} \Delta_i + \frac{\log^+(n\Delta_i^2)}{\Delta_i}\right)$$

for some C.

(b) From (a), we have $^\forall i \in \{1, \ldots, k\}$,

$$\Delta_i \mathbb{E}\left[T_i(n)\right] \leq C\left(\Delta_i + \frac{\log^+(n\Delta_i^2)}{\Delta_i}\right)$$

for some C. Let $\delta > 0$ be some small constant, which will be decided later. Then

$$\begin{aligned} R_n &\le \sum_{i:0<\Delta_i\le\delta} \Delta_i \mathbb{E}\left[T_i(n)\right] + \sum_{i:\Delta_i>\delta} \Delta_i \mathbb{E}\left[T_i(n)\right] \\ &\le n\delta + C\sum_{i:\Delta_i>\delta} \Delta_i + C\sum_{i:\Delta_i>\delta} \frac{\log^+(n\Delta_i^2)}{\Delta_i}. \end{aligned}$$

Let $\Delta_i = r\delta$ for Δ_i's with $\Delta_i > \delta$, where $r > 1$. Then

$$\frac{\log^+(n\Delta_i^2)}{\Delta_i} \le \frac{\log^+(n\delta^2) + 2\log r}{\delta r} \le \frac{\log^+(n\delta^2)}{\delta} + \frac{2}{\delta}$$

for Δ_i's with $\Delta_i > \delta$. Hence

$$R_n \le n\delta + C\sum_{i:\Delta_i>\delta} \Delta_i + \frac{Ck}{\delta}\left(\log^+(n\delta^2) + 2\right).$$

Choose $\delta = \sqrt{\frac{k\log^+ k}{n}}$. Then

$$\begin{aligned} R_n &\le \sqrt{kn\log^+ k} + \frac{Ck\sqrt{n}}{\sqrt{k\log^+ k}}\left(\log^+(k\log^+ k) + 2\right) + C\sum \Delta_i \\ &\le \sqrt{kn\log^+ k} + C\sqrt{kn}\frac{1}{\sqrt{\log^+ k}}\left(2\log^+ k + 2\right) + C\sum \Delta_i \\ &\le \sqrt{kn\log^+ k} + C\sqrt{kn\log^+ k} + \frac{2C\sqrt{kn}}{\sqrt{\log^+ k}} + C\sum \Delta_i \\ &\le C\left(\sqrt{kn\log^+ k} + \sum_{i=1}^{k} \Delta_i\right). \end{aligned}$$

Theorem (Cramér)**.** *Suppose $X_1, \ldots, X_n$ are i.i.d. random variables whose values lie in $\mathbb{R}^d$ with finite empirical mean defined by $S_n = \frac{1}{n}\sum_{i=1}^n X_i$. Define μ_n to be the law of S_n. Then μ_n satisfies a large deviation principle on $\mathbb{R}^d$ with rate function given by the Legendre transform of the cumulant generating function of X_1. That is,*

- *For any closed set $F \subset \mathbb{R}^d$,*
$$\limsup_{n\to\infty} \frac{1}{n}\log \mu_n(F) \le -\inf_{x\in F} \Lambda^*(x),$$
- *For any open set $G \subset \mathbb{R}^d$,*
$$\liminf_{n\to\infty} \frac{1}{n}\log \mu_n(G) \ge -\inf_{x\in G} \Lambda^*(x),$$

where $\Lambda^*(x) = \sup_{\lambda\in\mathbb{R}^d}\{\langle\lambda, x\rangle - \Lambda(\lambda)\} = \sup_{\lambda\in\mathbb{R}^d}\left\{\langle\lambda, x\rangle - \log\mathbb{E}\left[e^{\langle\lambda, X_1\rangle}\right]\right\}$.

Theorem (Gärtner-Ellis)**.** *Suppose $X_1, \ldots, X_n$ are random variables whose values lie in $\mathbb{R}^d$ with finite empirical mean defined by $S_n = \frac{1}{n}\sum_{i=1}^n X_i$. Define μ_n to be the law of S_n. Suppose the limit*

$$\Lambda(\lambda) := \lim_{n\to\infty} \frac{1}{n}\log\mathbb{E}\left[e^{n\langle\lambda, S_n\rangle}\right]$$

exists for all $\lambda \in \mathbb{R}^d$ and $0 \in \mathcal{D}_\Lambda^\circ$, where $\mathcal{D}_\Lambda$ is the effective domain of Λ. Then

- *For any closed set $F \subset \mathbb{R}^d$,*
$$\limsup_{n\to\infty} \frac{1}{n}\log \mu_n(F) \le -\inf_{x\in F} \Lambda^*(x),$$

- *For any open set* $G \subset \mathbb{R}^d$,
$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(G) \geq - \inf_{x \in G \cap \mathcal{F}} \Lambda^*(x),$$
where $\Lambda^*(x)$ *is the rate function given by the Legendre transform of* $\Lambda(\lambda)$*, and* $\mathcal{F}$ *is the set of exposed points of* $\Lambda^*(x)$*.*

Furthermore, if Λ *is essentially smooth and lower semicontinuous, then* $\mathcal{F}$ *can be replaced by the entire domain of* Λ^* *(i.e. LDP holds).*